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CS 499 - 4-1 Journal

Professor Phillips

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Part One:

Looking back at where I started in this program, my career goals have stayed the same, but my approach has changed completely. Coming from an IT Technician background, I've always been the guy fixing things after they break. My goal was to move from maintaining systems to building them. While I still want to be a software engineer, my focus has shifted from just writing code that works to engineering systems that are stable and efficient. It is one thing to build a search tool but it is another to ensure it does not crawl to a halt when the data gets messy. That is the kind of preventative maintenance mindset I want to bring to my development career.

I have spent some time researching what comes next, specifically looking into Systems Architecture and potentially pursuing a certification like the CSSLP or Certified Secure Software Lifecycle Professional. I want to bridge the gap between IT infrastructure and development. Right now, I feel confident that I have checked the boxes for Course Outcome 3 by designing a computing solution based on algorithmic principles while managing performance trade-offs. I have also achieved Course Outcome 4, since I am using advanced self-balancing techniques to ensure the software stays fast and meets the high performance standards expected in a real government or business environment. Outcomes related to security and professional communication remain my primary focus for the final stages of the capstone.

Checkpoint	Software Design and Engineering	Algorithms and Data Structures	Databases
Name of Artifact Used	Weight Tracking App (CS 360)	Binary Search Tree (CS 300)	Animal Shelter Project (CS 340)
Status of Initial Enhancement	Completed. Refactored code to the MVVM design pattern with LiveData.	Completed. Converted BST to a self-balancing AVL Tree.	Not started yet.
Submission Status	Milestone Two (Enhancement One) submitted.	Milestone Three (Enhancement Two) submitted.	Planned for the final milestone.
Status of Final Enhancement	Functionality is stable; architecture is fully modular.	Optimized memory and interactive search.	Plan includes creating a RESTful Web API using Flask.
Uploaded to ePortfolio	Code uploaded to GitHub; narrative finished.	Code uploaded to GitHub; narrative finished.	Pending implementation.

Status of Finalized ePortfolio	Ready for instructor review.	Ready for instructor review.	Pending.
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Part Two: Progress Update

Regarding the category of Software Design and Engineering, this artifact is fully complete. By applying the MVVM pattern to the Weight Tracking App, I have demonstrated the ability to separate concerns and create a professional, modular mobile application that is much easier to maintain than the original version.

For the Algorithms and Data Structure category, I have finished the transition from a standard Binary Search Tree to an AVL tree. This enhancement is now fully implemented and submitted, providing $O(\log n)$ search efficiency and ensuring that the tree remains balanced even with sorted data inputs. I have also finalized the recursive cleanup logic to ensure the system is free of memory leaks.

In the Database category, I am currently in the planning phase. My roadmap involves porting the local Python and MongoDB script into a Flask based REST API. This enhancement will focus on transitioning the code into a professional data service that incorporates a strict validation layer for all CRUD operations, ensuring that the system is both functional and secure against malformed data.

References

ISC2 CSSLP Certified Secure Software Lifecycle professional. ISC2 CSSLP Certified Secure Software Lifecycle Professional. (n.d.). <https://www.isc2.org/Certifications/CSSLP>